Spherical Easel Sprint Report 3

Prepared by: Wyatt Watson, Jason Bos, Samuel Rios, Ben Turner

Intended Progress

This sprint the team planned to finish the remaining tours and begin implementing some of the larger features required for the project. Each member was responsible for developing and completing their assigned tours. Alongside the tours, the team planned to begin implementing user action detection, completion condition flags, Firebase integration, and translation integration.

Week 1 was intended to focus on making user action detection functional, implementing completion condition flags, and beginning Firebase and translation integration.

Week 2 was intended to focus on completing the reusable dynamic tour system and finishing the more complex additional tours.

Wyatt: My goals for this sprint were to implement and solve user input detection and object interaction within tours. Review and improve file organization while resolving merge conflicts from each member's commits. Further improve the UI and address any existing usability issues. Investigate the translation feature and determine how it could be integrated into the existing tour system.

Sam: My primary goal for this sprint was to create my set of two tours for Spherical Easel. My two tours that I was to create were “Move an Object” and “Create a Segment Longer Than Pi”. I also wanted to do some extra research on a way to allow users to create tours via GUI.

Ben: My plan for this sprint was to build my two assigned tours for the Spherical Easel site. Those were an alternate “Create a Circle” tour and a “Measure a Line Segment” tour. Each tour needed its own way of detecting that the user had completed the action, so I planned on implementing that from our other tours and tried to look into how that could work for our baseline tour system. Also planned on looking into Firebase and translation options.

Jason: My goal for this sprint was to implement the “Create Point Behind Sphere” and “Alternate Segment Creation”. The goal was to make these tours advance based on user input. I also planned on testing for bugs if the user doesn’t do what the tours tell them to do.

Progress Reflection

Overall, the team made good progress toward the primary goal of completing the remaining tours. We were able to establish a working approach for detecting user actions and checking whether users had completed certain actions within a tour. This allowed us to begin developing more interactive tours that respond to what the user actually does within Spherical Easel rather than simply guiding them through a sequence of steps.

However, not all of the planned features were completed. While the team made progress on the tour system, Firebase integration and translation integration were not completed during this sprint. Additionally, there are still areas of the existing implementation that need refinement, particularly regarding UI responsiveness, testing, and reducing hardcoded functionality.

Wyatt: Initially, we had a little bit of a head start. The solution for detecting user input was to watch the Pinia stores for changes, so it mainly came down to developing the core functionality and having each member work on their assigned tours.

I was able to complete my assigned tours and implement the user input detection and object interaction logic needed to support them. This also gave us a better understanding of how we could use the existing application state to determine whether a user had successfully completed an action.

However, I did not have enough time to investigate UI testing as much as I wanted. There are still several issues and hardcoded features that need to be corrected. Additionally, we were not able to get to the Firebase features or translation integration. These features, along with further refinement of the tours, will carry over into the next sprint.

Sam: Fortunately, I was able to complete my set of tours with the help of Wyatt. I definitely should've managed my time a little more wisely and not set it all aside until the very last minute. Creating the tours themselves (after figuring out about Driver.js and looking at Wyatt's example for automation) was really fun to do, to see how I as a user would want to be guided.

Ben: I merged the team's development branch into mine since I was mostly working on the codebase locally up to this point, and wrote both of the new tour functions. The alternate circle tour takes the user to the Create Circle tool and finishes when a new circle is added to the store. And the measure tour takes the user to the Measurement Tools section and finishes when a new segment measurement is created. Both of these weren't too bad to implement, and I had fun trying to work with the measuring tool specifically. Also got a better idea of how Driver.js worked which was nice.

Jason: I finished the "Create Point Behind Sphere" tour, which took most of my time. We decided that we wouldn't need to make a tour for the other way to create a segment. I spent a little time looking at testing for bugs, but Driver.js does a good job of not handling users doing unexpected things, so I didn't find any problems in my limited testing.

Problems Encountered

One challenge encountered was underestimating the amount of work required to develop the logic behind each tour. While the overall structure of the tours was established, each tour required its own custom logic to detect user actions and determine when certain conditions had been met. This took more time than initially anticipated.

Additionally, with midterms taking place during this sprint, time was particularly limited for all members. Coordinating the development of individual tours and waiting for each member's implementation to be completed also introduced some delays. This made it difficult to dedicate enough time to the additional features we originally planned.

Wyatt: Time was the biggest issue for the project this sprint. Luckily, I already knew how to access and work with the Pinia stores, but tying together the custom logic and Driver.js functionality took considerable time. Each tour needed to interact with the application differently, meaning that we couldn't simply reuse the same logic for every tour.

Additionally, coordinating the development of each tour and making sure everything worked together took longer than expected. This left less time for testing, UI improvements, and investigating the other planned features. Moving forward, we need to account for the time required to integrate everyone's work rather than only considering the time needed to develop each individual feature.

Sam: One of the largest problems actually occurred outside of the actual project. Having fall break land right on this was very unfortunate (but also fortunate in a way). I was invited by my friends from MTU to go see them over break and I had not seen them in a long time, so I did. The trip cut into a lot of my sprint time and I did not have a chance to work on it during that weekend. Then, I ended up getting sick from them, resulting in even less time to work on the project. Other than my trip and illness, there weren’t many issues with the actual development of my tours.

Ben: Most of the problems I encountered were setting up the entire project on a new laptop since my old one’s fan broke making it almost impossible to use. It wasn’t that bad, just very annoying since I had to install everything for the project again and I ran into some problems setting it up too. Once I got there, the tours went pretty smoothly outside of a few Firebase errors. I had some distractions here and there because of fall break and life in general, and because of that I didn’t get a definitive answer on how to implement Firebase for saved tours exactly, but that’s something Wyatt’s getting clarification on in a few days anyways. The same goes for translations.

Jason: The main problem I ran into was figuring out how to detect when the user places a point behind the sphere. I couldn’t figure out where the coordinates of the points were stored on the sphere, so I just detected if the user held Shift while a point was created. This is a lot simpler than finding if a point was created on the other side of the sphere.

Projected Progress

Although we did not complete everything planned for Sprint 3, a significant amount of critical work was accomplished. The team now has a much more developed tour system, with most of the assigned tours completed and a better understanding of how to handle user interactions and completion conditions.

Sprint 4 will involve catching up on the remaining features and refining the work completed throughout the semester. This will be a particularly important sprint, as we need to ensure that all sponsor requirements are satisfied and that the project meets the expected professional standards. Our focus will shift toward completing the remaining integrations, improving reliability, and making the overall experience more polished.

Wyatt: I will be working with the other members to finish the Firebase system, translation system, and final responsive Tour-Tab UI. The tours are largely completed at this point, so we need to focus on refining them and addressing the remaining issues.

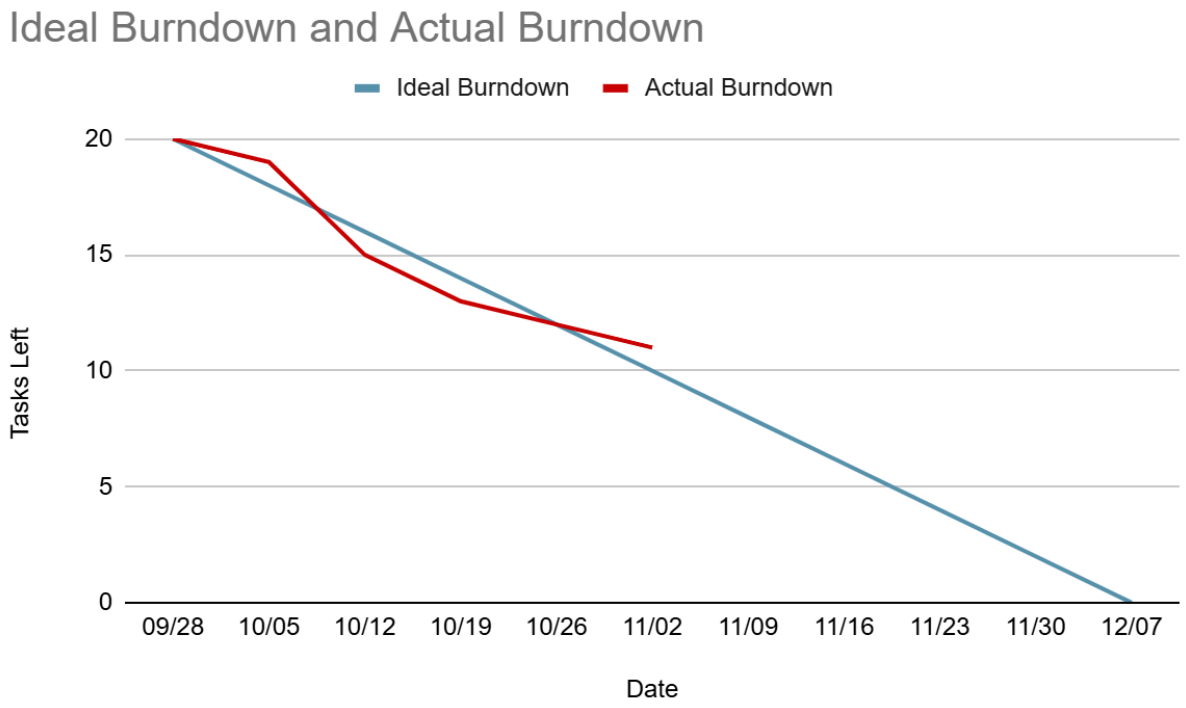
I also want to spend more time reviewing the existing implementation, reducing hardcoded functionality, and making sure that the different tours work consistently. Testing will be especially important, as we need to make sure that the tours function correctly across different user interactions and screen sizes. With the remaining features completed, we should be in a good position to finalize the project and prepare it for our sponsors.

Sam: I will most likely begin working on documentation for the project, since I have already created the id's for the html objects for Driver to use. I will also let family members test out our tours and see if they find it intuitive or not. Not only that, but I will try and see if I can implement the translation feature that they have onto our tours.

Ben: In Sprint 4 I will refine the tours I worked on and shift focus to Firebase and how to save completed tours there effectively. Along with helping with documentation and translations once our team settles on how to approach those things. This coming week's meeting will hopefully answer just that along with any other questions us or the sponsors may have.

Jason: In the next sprint, I plan on helping to fix any problems that the tours might have, continue to test more thoroughly, and help work on documentation. I also plan on working on the translations because we haven't implemented that yet.

Update of Burn Down Chart



Teamwork Reflections

It was a tight sprint, but the team worked well together communicating and distributing the workload. Despite the time constraints and midterms, everyone was able to contribute to the project and make progress on their assigned responsibilities. We also did a good job helping one another when issues came up, particularly when working through the custom logic required for individual tours.

Wyatt: Very proud of the team. It was a tough sprint, and we worked well together. Communication was good, and everyone was able to contribute to getting the tours completed. We also did a good job helping one another when we ran into problems, which was especially important given the time constraints.

One thing we still need to improve is making sure that we communicate delays and potential problems as early as possible. With the amount of integration required between everyone's work, even small delays can affect the rest of the team. If we can maintain our current level of communication while improving how early we address these issues, I believe we will be able to get everything done in Sprint 4, even with the additional work we need to catch up on.

Sam: This past sprint (even with fall break) had good communication between the four of us! Wyatt kept us in check and he was helping us out in case we needed it. We all made sure we communicated when we had finished our parts. I honestly should've mentioned it to the team earlier that I had gotten sick and I was not able to work on the tours sooner, but I was too ashamed to say anything. I will be sure to communicate asap from now on.

Ben: The team communicated better this spring than the last one, and I think fall break helped ease some of the urgency with the workflow. Wyatt was especially good with keeping us in the loop with what needs to be done and when which was very helpful. I think I did well with communication personally even though I might need to try and get on top of things a little earlier.

Jason: This sprint was the most work-intensive so far, so we needed to communicate well. I think we did a good job of that. When I got stuck on something, I let the team know and they helped me. I could've done a little better at letting the team know about my progress with the sprint, but other than that, I think we did a good job getting everything done during a busy time of the semester.

Conclusion

Overall, Sprint 3 was a productive sprint for the team, even though we were unable to complete all of our original goals. The majority of the assigned tours have been completed, and we have made significant progress on the underlying functionality that allows tours to respond to user actions. This is an important step toward having a reusable and dynamic tour system.

There are still several features that need to be completed, including Firebase integration, translation integration, and further UI refinement. Sprint 4 will be important for bringing all of these pieces together, resolving remaining issues, and ensuring that the final project meets our sponsors' expectations.

Wyatt: I feel that the team is in a good position going into our final sprint. We have made significant progress on the core functionality of the project, and the majority of the tours are now in place. While there is still work to be done, we have a much better understanding of how the system works and what needs to be completed.

My biggest concern going into Sprint 4 is making sure that we have enough time to refine and test everything rather than simply focusing on getting the remaining features implemented. We need to make sure that the project is not only functional but also reliable and intuitive for users. If we can stay organized and maintain our communication, I believe we can finish the remaining work and deliver a project that meets our sponsors' expectations.

Sam: I feel really good about how our project is looking. We have made great progress as a team and the tours all seem to be working the way they should. I feel confident that we will be able to complete the next sprint and to finish up the requests that our sponsors have given us.

Ben: I feel good where this team is at right now for the final sprints. Most of the tours are now implemented and the detection approach works well, so the remaining work is perfecting them along with implementing Firebase and translations. And of course more user testing. I'm confident that we'll get there eventually and our sponsors will be happy with our progress.

Jason: I think we're in a good spot; we have most of the work done, and now it's just time to finalize everything and fix any bugs. This still will require a decent amount of work, but I think we are doing great, and we won't have to rush to get everything done in time.